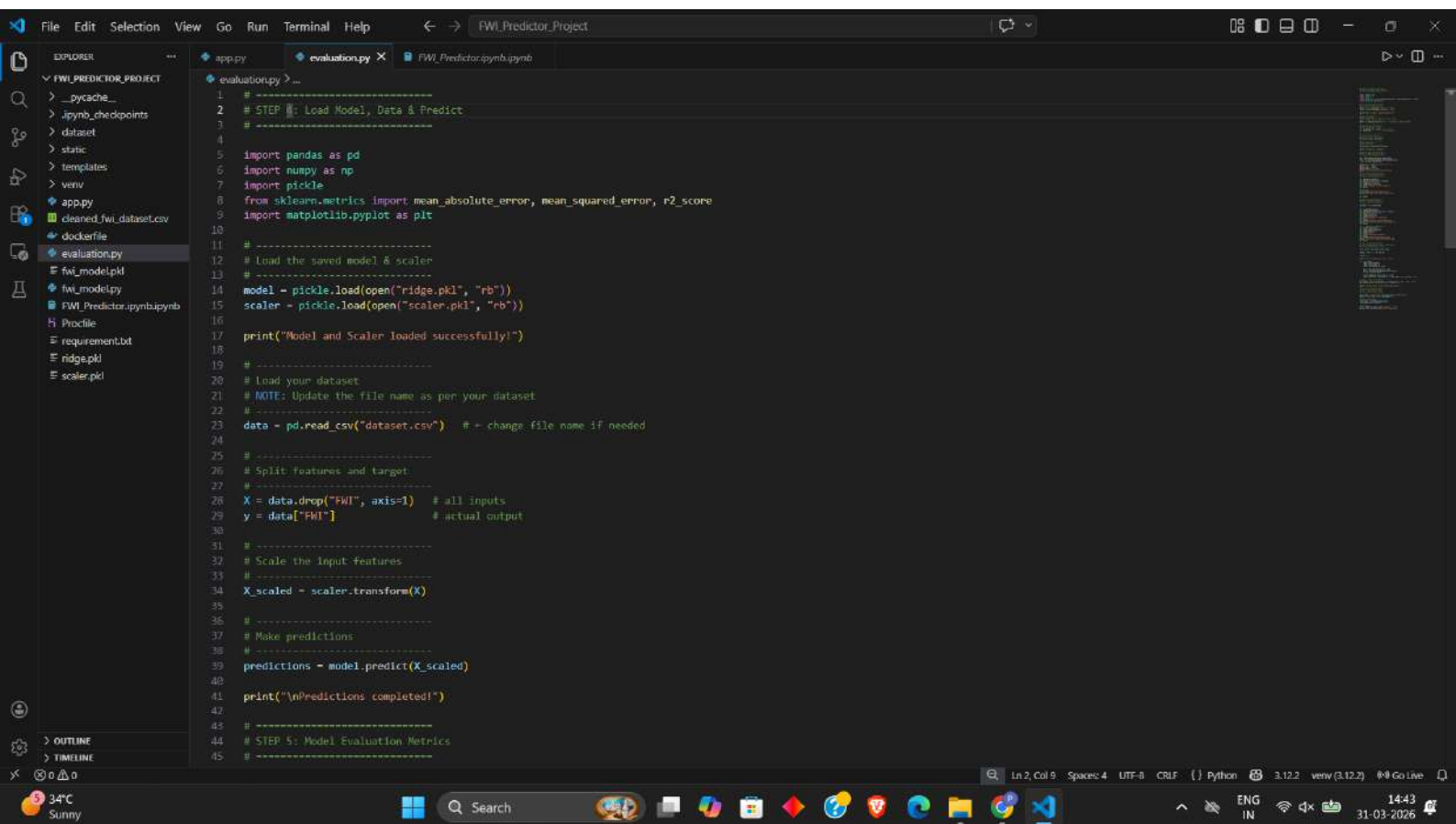
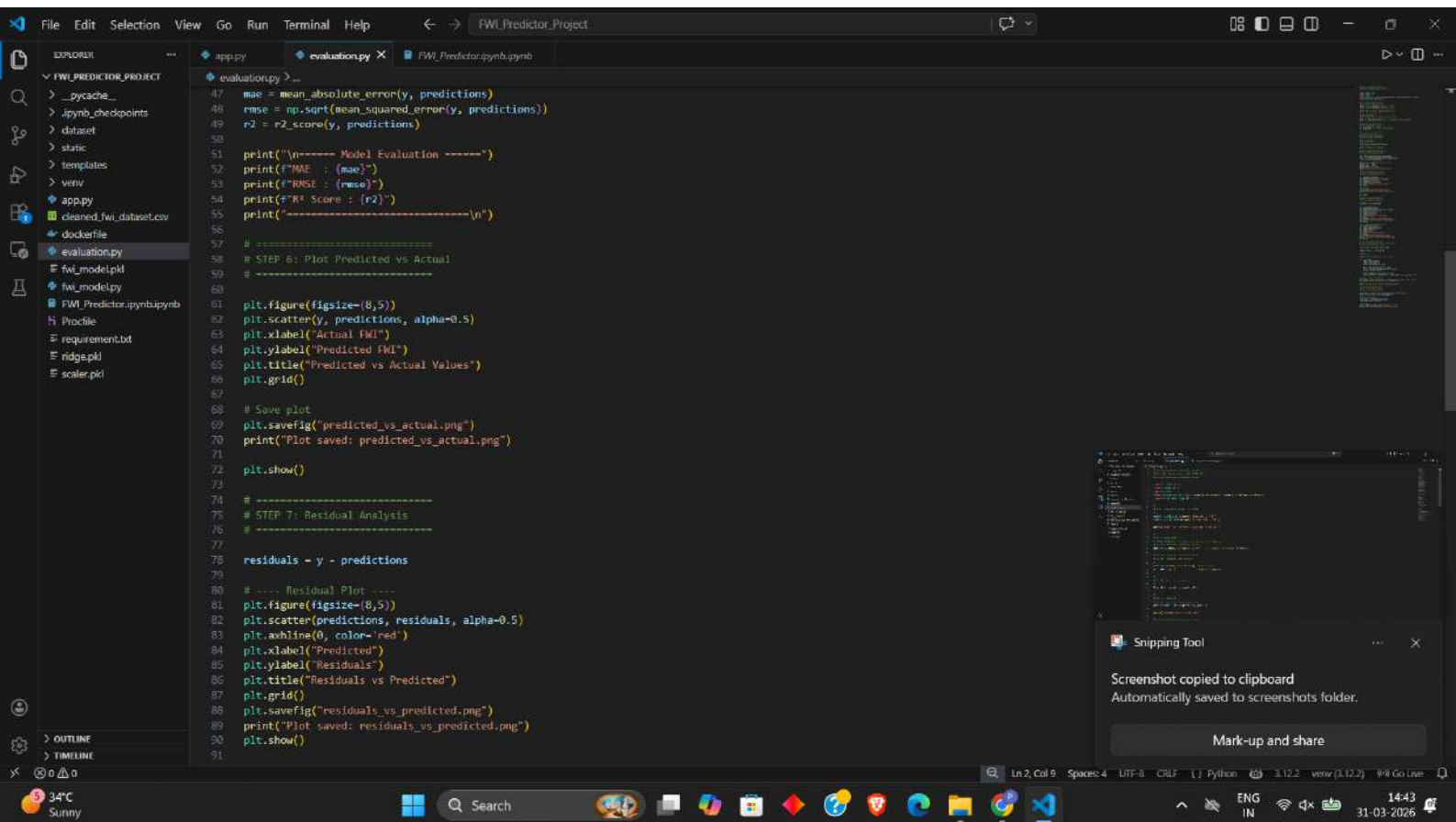






File Edit Selection View Go Run Terminal Help FwI_Predictor_Project

EXPLORER

- FWI_PREDICTOR_PROJECT
 - __pycache__
 - __ipynb_checkpoints
 - dataset
 - static
 - templates
 - venv
 - app.py
 - cleaned_fwI_dataset.csv
 - dockerfile
 - evaluation.py
 - fwI_model.pkl
 - fwI_model.py
 - FWI_Predictor.py:nbipynb
 - Profile
 - requirements.txt
 - ridge.pkl
 - scaler.pkl

evaluation.py

```
93 plt.figure(figsize=(8,5))
94 plt.hist(residuals, bins=30)
95 plt.xlabel("Residual")
96 plt.ylabel("Count")
97 plt.title("Residual Distribution")
98 plt.grid()
99 plt.savefig("residual_distribution.png")
100 print("Plot saved: residual_distribution.png")
101 plt.show()
102
103 # =====
104 # STEP 8: Hyperparameter Tuning (Ridge alpha)
105 # =====
106
107 from sklearn.linear_model import Ridge
108
109 alphas = [0.1, 1, 5, 10, 20, 30]
110
111 results = []
112
113 print("\n----- Ridge Alpha Tuning -----")
114
115 for a in alphas:
116     ridge = Ridge(alpha=a)
117     ridge.fit(X_scaled, y)
118     pred = ridge.predict(X_scaled)
119
120     mae_a = mean_absolute_error(y, pred)
121     rmse_a = np.sqrt(mean_squared_error(y, pred))
122     r2_a = r2_score(y, pred)
123
124     results.append([a, mae_a, rmse_a, r2_a])
125     print(f"Alpha={a} - MAE={mae_a:.4f}, RMSE={rmse_a:.4f}, R2={r2_a:.4f}")
126
127 # Convert results to DataFrame
128 df_results = pd.DataFrame(results, columns=["Alpha", "MAE", "RMSE", "R2"])
129 df_results.to_csv("tuning_results.csv", index=False)
130
131 print("\nTuning results saved: tuning_results.csv\n")
132
133 # =====
134 # STEP 9: Save the Best Model
135 # =====
136
137 best_alpha = df_results.sort_values(by="MAE").iloc[0]["Alpha"]
```

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

Ln 2, Col 9 Spaces: 4 UTF-8 CRLF Python 3.12.2 venv (3.12.2) IP: Go Live

34°C Sunny Search 14:43 31-03-2026

Model training completed

Evaluation Metrics:

MAE: 0.4769024031723191

RMSE: 0.7513475825725587

R2 Score: 0.9813740484635791

Alpha Tuning Results:

Alpha: 0.1 R2: 0.9865415746737278

Alpha: 1 R2: 0.9813740484635791

Alpha: 10 R2: 0.9677193576959423

Model and scaler saved successfully!

(venv) PS C:\Users\Roshani\OneDrive\Documents\FWI_Predictor_Project>

